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## Introduction to Cryptography

A1. For the first lab (1.1), I first began by using the ZYWINo1 VM and used CMD then OpenSSL to generate Pasha's RSA key pair which was created with a 4096-bit private key stored in the directory C:\Users\Pasha\privateKey with the public key in the same directory as well but named \publicKey. As the recipient, Pasha was able to keep the private key confidential while using the other one for sharing. This is how Pasha's public key was sent to Rina and saved on the ZYWINo2 VM in the directory C:\Users\Rina. While on the ZYWINo2 VM, I encrypted the confidential file within C:\Users\Rina\creditCardInfo.txt by using the public key created by Pasha. This was later saved and encrypted using Pasha's public key, but the only way to decrypt it is by using the private key. Since the file emailed to Pasha was encrypted, this protects the attachment as it's being transmitted across the network. I then decrypted creditCardInfoEncrypted.txt by using Pasha's private key to confirm the recovered creditCardInfo.txt within C:\Users\Pasha. Once I was able to make sure it was restored properly, I completed the exercise and deleted the VMs.

A2. The first challenge I had faced when trying to send the encrypted message was when I was trying to confirm if the public key Rina had gotten was actually belonging to Pasha. Since the public key was sent by email an attacker could have been able to intercept it and used their own public key which would've made Rina unable to encrypt the credit card file the correct way. To verify this you could use a fingerprint via a separate channel which is trusted or by using a Public Key Infrastructure (PKI) that contains a certificate of the key that belongs to Pasha.

The second challenge I had noticed is that RSA was only able to encrypt data that was smaller than the key length so it only worked with 4096-bit public key because the creditCardInfo.txt is small enough but anything larger than that would exceed the capable limit and fail. The solution would be to encrypt using a fast algorithm such as AES which would be able to handle those sizes. So in conclusion, you could still use RSA to encrypt the small AES key while AES encrypts the larger file, so the two work together to handle any file sizes.

B1. The ZYWINo1 was the first step of the lab and I used OpenSSL to create a random key for the AES-256 encryption which I saved as finance.key within the C:\ProgramData\EncryptionKeys directory. This was the only secret key that I used for both of

the encryption and decryption keys for the financial report which is what defines symmetric encryption (zyBooks, 2025). I also transferred the finance.key over to the other VM (ZYWINo2) by using scp which was sent and saved to the directory C:\ProgramData\EncryptionKeys within that machine. The reason for doing this was to ensure that the key was never exposed with the data it's meant to protect, which is why I used SCP. I also made Rina have read-only permission for the finance.key file which enforces the access controls rule and it means that Rina can decrypt the file but can't overwrite any of its contents, the only thing Rina can do is read it. On ZYWINo1, I also encrypted the financial report within C:\Users\Pasha\Documents\financialReport.xlsx with the finance.key file by using the same AES-256 encryption format which made the spreadsheet turn from readable into unreadable ciphertext. Then I emailed financialReport.xlsx.enc over to Rina and it was saved on the ZYWINo2 VM, which shows that even while the file was transmitted across the network, the report stayed protected the entire time. Lastly, on the ZYWINo2 VM I decrypted the financialReport.xlsx.enc using the same finance.key, saving the recovered file as financialReport.xlsx within C:\Users\Rina\Documents, and the decrypted spreadsheet matched the original.

B2. The first challenge that I had faced involved securely delivering the shared key. Since the same single key is used for both encryption and decryption, the finance.key must reach Rina without being intercepted during transmission. If an attacker had intercepted this they would've been able to decrypt the financial report without needing anything else other than the file itself. So the solution for this would've been to move the key into a secure channel rather than sending it over with the file through email. This is why SCP was the best solution for this scenario because it encrypts the transfer and keeps things confidential.

The second challenge I had noticed was that once the key had reached Rina's machine, anyone could read or overwrite the finance.key file and have access to financial information. The solution for this would be to create a read-only access permission so that even if the file was decrypted the only access a user could do would be to read the report. This restriction of access aligns with confidentiality and integrity of data protection.

## **References**

zyBooks. (2025). D830: Introduction to cryptography. Western Governors University.